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Odds

Niche prediction markets
for African communities

Built on BOT Chain testnet (chain ID 968)
Prepared for the BOT Chain Africa Builder Challenge

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1. Abstract

Odds is a decentralized prediction market for BOT Chain that focuses on niche questions: local football results, regional events, weather, crypto milestones and other topics where a community often knows more than a global market does. Participants back YES or NO with the chain's native token. When the market closes, a designated resolver reports the outcome and everyone who backed the winning side shares the entire pool, less a 2% protocol fee, in proportion to their stake. If a market cannot be settled fairly, everyone is refunded.

This paper describes the market mechanism, the settlement and payout rules, the contract architecture, the trust assumptions around resolution, and the planned path toward oracle-backed markets.

2. The Problem

Prediction markets aggregate dispersed knowledge into a price. Global platforms do this well for headline events but largely ignore the questions that matter most to specific communities, and they carry costs and access barriers that exclude many users.

- **Coverage gaps.** Questions about African leagues, local elections, regional weather or small crypto events rarely attract enough liquidity on global venues to be listed at all.
- **Heavy mechanics.** Order books and automated market makers need liquidity providers, pricing curves and active management, which is difficult to bootstrap for a niche topic.
- **Opaque settlement.** Users must trust a platform's internal process for deciding outcomes and moving money.

3. The Odds Approach

Odds uses the simplest mechanism that works at small scale: a parimutuel pool. There is no order book, no liquidity provider and no pricing curve. Everyone bets into the same two pools and the pools themselves define the odds.

- **One pool per side.** Each market has a YES pool and a NO pool. The implied probability of YES is its share of the combined pool.
- **Winners share everything.** After resolution, the whole combined pool, minus the fee, is divided among holders of the winning side in proportion to their stake.
- **No bootstrapping problem.** A market with a single bet on each side is already fully functional, which suits niche topics with few participants.
- **Settlement on-chain.** Stakes, resolution and payouts are all enforced by the contract.

4. Market Lifecycle

Every market is in exactly one of four states, derived from time and the resolution record.

State	When	Allowed actions
Open	Before the closing time	Anyone can bet YES or NO.

State	When	Allowed actions
Awaiting result	After closing, up to 7 days	The resolver reports the outcome. Betting is closed.
Resolved	After the resolver reports	Winners claim their share.
Void	No bets on the winning side, or no result within 7 days	Everyone claims a full refund of their stake.

Creating a market

Any user can create a market by providing a question, a closing time in the future and a resolver address. The question text is stored on-chain. Because the contract cannot judge a question, clear wording and resolution criteria are the creator's responsibility.

Betting

While a market is open, any user can place a bet on YES or NO with a native-token value. Stakes accumulate per address and per side, and one address may hold stakes on both sides.

Resolution

After the closing time, only the resolver can report the outcome. If nobody staked on the winning side, the market is voided instead of resolved, so that no funds are stranded and no fee is taken.

Claiming

Once a market is Resolved or Void, each participant calls `claim` once. In a Resolved market the contract pays winners; losers receive nothing. In a Void market it returns the participant's total stake on both sides.

5. Pricing and Payout Mathematics

```
impliedProbability(YES) = yesPool / (yesPool + noPool)
fee            = total - total * (10000 - 200) / 10000      (2%, taken at resolution)
pot            = total * 9800 / 10000
payout(user)   = userStakeOnWinner * pot / winningPool
```

A worked example makes the mechanics concrete.

Item	Value
YES pool	6 units
NO pool	4 units
Total pool	10 units
Implied probability of YES	60%
Fee (2%)	0.2 units
Pot to winners	9.8 units
Bettor with 3 units on YES, if YES wins	3 x 9.8 / 6 = 4.9 units

Item	Value
Same bettor if NO wins	0

> Parimutuel odds are final only when betting closes. A later bet shifts the pools and therefore the return for everyone on the same side, so the return a user sees before betting is an estimate. The Odds interface shows this estimate and labels it as such.

6. Smart Contract Design

The protocol is a single contract, `NichePredictionMarket`. It stores an array of markets plus mappings for YES stakes, NO stakes and claim records. The deployer is the owner, whose only privilege is withdrawing accrued fees.

Function	Access	Purpose
createMarket	Anyone	Create a market with a question, closing time and resolver.
bet	Anyone, while open	Stake on YES or NO.
resolve	Resolver only, after closing	Report the outcome, or void if the winning side is empty.
claim	Participants, when settled	Withdraw winnings or a refund, once.
withdrawFees	Owner	Collect accrued protocol fees.
status / getMarket	View	Read the derived state and market data.

Constants

Constant	Value	Meaning
FEE_BPS	200	Protocol fee of 2% on resolved pools only.
VOID_AFTER	7 days	Grace period after closing before an unresolved market refunds.

Events and safety patterns

- Events `MarketCreated`, `Bet`, `Resolved` and `Claimed` allow indexers to rebuild any market.
- A claim is marked as taken before funds are sent, preventing double withdrawal and re-entrancy.
- The fee is accrued only when a market is genuinely resolved. Voided and expired markets refund stakes in full.
- Betting is impossible after the closing time and resolution is impossible before it.

7. Resolution and Trust Model

Resolution is the one place where Odds depends on trust. The contract enforces the rules, but it cannot observe a football match or a rainfall total. A designated resolver address reports the result. That address can be an individual, a multisig wallet representing a community, or in future an adapter contract that reads from an oracle.

- **Resolver honesty.** A dishonest resolver can misreport. Users should check who the resolver is before betting; the address is displayed on every market page.
- **Resolver liveness.** If the resolver does not act within 7 days of closing, the market voids and all stakes are refundable, so a silent resolver cannot lock funds.
- **Creator as resolver.** A creator may name themselves. Users should treat such markets with caution unless they trust the creator.
- **Question ambiguity.** Poorly worded questions invite disputes the contract cannot settle. Resolution criteria should be unambiguous and ideally linked to a public data source.

8. Application Layer

The reference application is a static site with no backend. It provides a landing page, a market browser filterable by state, a market page with a live odds bar, return estimator, betting controls, resolver controls and claim button, and a market creation form. Wallet connection prompts users to switch to or add BOT Chain testnet, and all reads come from the public RPC endpoint.

9. Deployment on BOT Chain

Parameter	Value
Network	BOT Chain testnet
Chain ID	968
RPC endpoint	https://rpc.bohr.life
Block explorer	https://scan.bohr.life
Contract address	Published at deployment
Language / compiler	Solidity 0.8.24 (checked on solc 0.8.26)

10. Risks and Limitations

- **Not audited.** The contract is a reference implementation and has not been independently reviewed.
- **Late-bet advantage.** Because odds are determined at close, users who bet last see the most information. This is inherent to parimutuel pools.
- **No early exit.** Positions cannot be sold or withdrawn before settlement.
- **Rounding dust.** Integer division can leave a negligible remainder in the contract after all claims.

- **Legal and regulatory.** Prediction markets are regulated differently across jurisdictions. Operators and users are responsible for compliance with local law.
- **Native token only.** Pools are denominated in the chain's native token and exposed to price volatility.

11. Roadmap

Stage	Focus
Next	Independent audit; fuzz-tested payout invariants (total paid out never exceeds the pot); verified source on the explorer.
Then	Oracle-backed resolution adapters (for example Chainlink or UMA) so resolvers need not be trusted individuals; dispute window before final settlement.
Later	Stablecoin markets; market categories and curated feeds for African sports and events; an optional automated market maker for markets that want continuous trading.

12. Conclusion

Odds trades sophistication for accessibility. A parimutuel pool needs no liquidity providers, produces readable odds and settles entirely on-chain, which makes it well suited to the niche, community-driven questions that global markets overlook. As resolution moves from trusted resolvers toward oracles, Odds can grow from a community tool into infrastructure for local information markets.

Disclaimer

> This document describes software that is experimental, deployed on a test network and has not been independently audited. Nothing here is financial, legal or investment advice, and nothing here promises returns or the availability of any feature. Test-network tokens have no monetary value. Roadmap items are intentions, not commitments.